

QUESTIONS

Q1
Student Registration System
10 marks

Q2
E-Commerce Cart System
10 marks

Q3
Library Management System
10 marks

Q4
Employee Data Processing
10 marks

Q5
Online Voting System
10 marks

5/5 answered

Q1 Student Registration System

10 marks

Suppose that a university system stores student data, where each student must be unique and can enroll in multiple courses.

- Enumerate three Collection interfaces suitable for this system.
- Design a Collection structure (diagram/representation) to store students and their enrolled courses.
- Write a Java program using HashSet and HashMap with Generics to:
 - Add students
 - Map students to courses
 - Display data using Iterator

Your Code

```
1 import java.util.*;
2 public class University{
3     public static void main (String[] args){
4
5         HashSet<String>students = new HashSet<>();
6         HashMap<String, ArrayList<String>>map = new HashMap<>();
7
8         students.add("Anjali");
9         students.add("Aishma");
10
11         ArrayList<String>c1 = new ArrayList<>();
12         c1.add("Java");
13         c1.add("C++");
14     }
15 }
```

Input

stdin input

Output

Aishma -> [Python, DBMS]
Anjali -> [Java, c++]

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Q2 E-Commerce Cart System

10 marks

Suppose that an online shopping system maintains a cart where products are added, removed, and displayed in order.

- List three Collection classes applicable for cart management.
- Represent how cart items are stored using a List-based structure.
- Write a Java program using ArrayList and Iterator to:
 - Add items
 - Remove an item
 - Display all items

Your Code

```
1 import java.util.*;
2 public class Main {
3     public static void main(String[] args) {
4
5         ArrayList<String> cart = new ArrayList<String>();
6         Scanner sc = new Scanner(System.in);
7         for(int i = 0; i < 3; i++) {
8             cart.add(sc.next());
9         }
10        System.out.println("Cart Items:");
11        Iterator<String> it = cart.iterator();
12        int count = 0;
13        while(it.hasNext()) {
```

Input

Dress
Facewash
Sunscreen

Output

Cart Items:
Facewash
Sunscreen

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Q3 Library Management System

10 marks

Suppose that a library stores books with unique ISBN numbers and needs fast retrieval.

- Name three Map implementations in Java.
- Design a schema-like representation using Map for storing ISBN and book details.
- Write a Java program using HashMap to:
 - Insert book details
 - Retrieve a book
 - Display all books

Your Code

```
1 import java.util.*;
2 public class Library{
3     public static void main(String[] args) {
4         HashMap<Integer, String> books = new HashMap<>();
5
6         books.put(101, "Java");
7         books.put(102, "Python");
8         books.put(103, "C++");
9         System.out.println("Book: " + books.get(102));
10        System.out.println("All Books: ");
11        for(Map.Entry<Integer, String> b : books.entrySet())
12        {
13            System.out.println(b.getKey() + " - " + b.getValue());
14        }
15    }
16 }
```

Input

stdin input

Output

```
Book: Python
All Books:
101 - Java
102 - Python
103 - C++
```

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5/5 answered

Q4 Employee Data Processing

10 marks

Suppose that a company processes employee records and filters employees based on salary.

- (a) List three advantages of Generics in Collections.
- (b) Represent how employee data is stored using List with Generics.
- (c) Write a Java program using Iterator to:
 - Store employee objects
 - Display employees with salary > 50,000

Your Code

```
1 import java.util.*;  
2 class Main {  
3     public static void main(String[] args) {  
4         ArrayList<String> name = new ArrayList<String>();  
5         ArrayList<Integer> salary = new ArrayList<Integer>();  
6  
7         name.add("Anjali");  
8         salary.add(50000);  
9         name.add("Aishma");  
10        salary.add(60000);  
11        name.add("Abi");  
12        salary.add(70000);  
13  
14        Iterator<String> it1 = name.iterator();  
15        Iterator<Integer> it2 = salary.iterator();
```

Input

stdin input

Output

```
Aishma 60000  
Abi 70000
```

CO2-AT1-Scenario Based Programs

← Back

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Q5 Online Voting System

10 marks

Suppose that an online voting system ensures one vote per user and counts votes efficiently.

- Identify three suitable Collection types for this system.
- Design a structure using Set and Map for voters and vote counting.
- Write a Java program using HashSet and HashMap to:
 - Record votes
 - Prevent duplicate voting
 - Display results

Your Code

```
1 import java.util.*;
2 public class Main {
3     public static void main(String[] args) {
4         HashSet<String> voters = new HashSet<String>();
5         HashMap<String, Integer> votes = new HashMap<String, Integer>();
6
7         String[] voter = {"anju", "kripa", "aish", "lisa"};
8         String[] candidate = {"A", "B", "A", "A"};
9
10        for (int i = 0; i < voter.length; i++) {
11            if (!voters.contains(voter[i])) {
```

Input

stdin input

Output

{A=3, B=1}